HOTEL REVENUE ANALYTICS

Hotel Pickup Pace Workflow

How the data was handled, how it was analysed, and why each method was chosen

1 Load	2 Build dates	3 Filter	4 Aggregate	5 Compare	6 Interpret
119,390 rows, 32 columns	arrival and booking date	Check-Out bookings only	by arrival date and lead time	hotel, weekday, segment, ADR	charts, shares and blocks

This document records the workflow behind the Hotel Booking Pace and Pickup Analysis report. Part 1 covers how the data was prepared, Part 2 how it was analysed, and each step states the reason it was done that way. The analysis uses the Hotel Booking Demand dataset (Antonio, Almeida and Nunes, 2019) for a City Hotel and a Resort Hotel, with arrival dates from July 2015 to August 2017.

Part 1. How the data was addressed

Step	What was done	Why
Load and inspect	Read the bookings CSV into pandas and checked the first rows, columns and shape: 119,390 bookings and 32 columns. The recomputation dataset matched the notebook's counts exactly.	Confirms the data is complete and that every figure in the report can be traced to the same source.
Build the arrival date	Combined arrival year, month and day into one datetime column.	The dataset stores the date in three separate columns. A single date is needed to group bookings by arrival and to read weekday and month.
Derive the booking date	Booking date = arrival date minus lead time (in days).	The dataset has no booking timestamp. Subtracting lead time is the way to place each booking on the calendar. By construction no booking falls after its arrival date.
Keep stayed bookings only	Kept reservation status Check-Out: 75,166 of 119,390 bookings. Removed 43,017 cancellations and 1,207 no-shows.	Pace and pickup describe demand that became a stay. Including bookings that never produced a stay would blend two different things.
Define lead time	Lead time is the number of days between booking date and arrival date (0 to 737 days).	It is the common axis for every pace curve, histogram and heatmap, so all views measure demand relative to arrival.
Define the groups	Hotel type as recorded. Weekend arrivals are Friday to Sunday and weekday arrivals Monday to Thursday. Market segment as recorded.	These are the splits a revenue manager actually forecasts and prices by, so each group gets its own pace profile.

Scope. All results describe bookings that were honoured (Check-Out). Booking dates are derived from lead time, not read from a booking system. Counts are booking records, not rooms.

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Part 2. How the data was analysed, and why

Analysis	How it was done	Why this way
Booking window	Histogram of lead time (50 bins) with median and mean, plus a cumulative share curve for 0 to 365 days.	Shows the whole shape of the booking window. The median (45 days) is reported next to the mean (80 days) because lead time is strongly right-skewed.
Pace curve of one date	For the busiest arrival date, bookings were counted by lead time and accumulated from the longest lead time to arrival.	Shows how demand builds toward arrival for a single date and exposes the steps that large bookings create.
Pickup heatmap	Bookings per arrival date and lead time (0 to 120 days) for August 2017, with daily totals alongside. Colour scale capped at 10.	One row per arrival date reveals date-specific pickup. The cap keeps ordinary cells readable while still marking the densest ones.
Hotel, weekday and segment comparison	Each group's lead-time distribution was shown in raw counts and as a cumulative share of the group's own bookings. Medians and share of bookings within 7 days, same-day and over 180 days were tabulated.	Groups differ in size: City holds 61.5% of stayed bookings and weekdays cover four days against three. Raw counts mostly show volume, while shares isolate timing.
ADR by lead time	Average ADR was calculated for lead-time buckets (same day, 1 to 7, 8 to 30, 31 to 60, 61 to 90, 91 to 180, 181 to 365 and 366+ days), by hotel and overall. A sum-of-ADR pace curve was kept for reference.	The sum of ADR by lead time mirrors booking volume and cannot show value. Average ADR per bucket isolates whether late or early bookings carry different rates.
Top 10 arrival dates	For the ten busiest dates, bookings were plotted by lead time on separate panels. The largest single lead-time spike was traced to its market segment, hotel and share of the date's bookings.	Separate panels give each date its own scale so single-day blocks stay visible. Tracing each block shows what produced the peak.

Definitions

Term	Meaning in this analysis
Lead time	Days between the booking date and the arrival date.
Pickup	Bookings made at a given lead time for a given arrival date.
Pace curve	Bookings by lead time, or accumulated toward arrival, showing how demand builds.
ADR	Average daily rate recorded on each booking.
Block	A large number of bookings for one arrival date made on a single day, identified as the largest single lead-time spike.

Tools. Python with pandas for data preparation and aggregation, and matplotlib for the charts. Every statistic in the report was calculated from the data, not estimated from the charts.

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